

EarthBridge: A Solution for 4th Multi-modal Aerial View Image Challenge Translation Track

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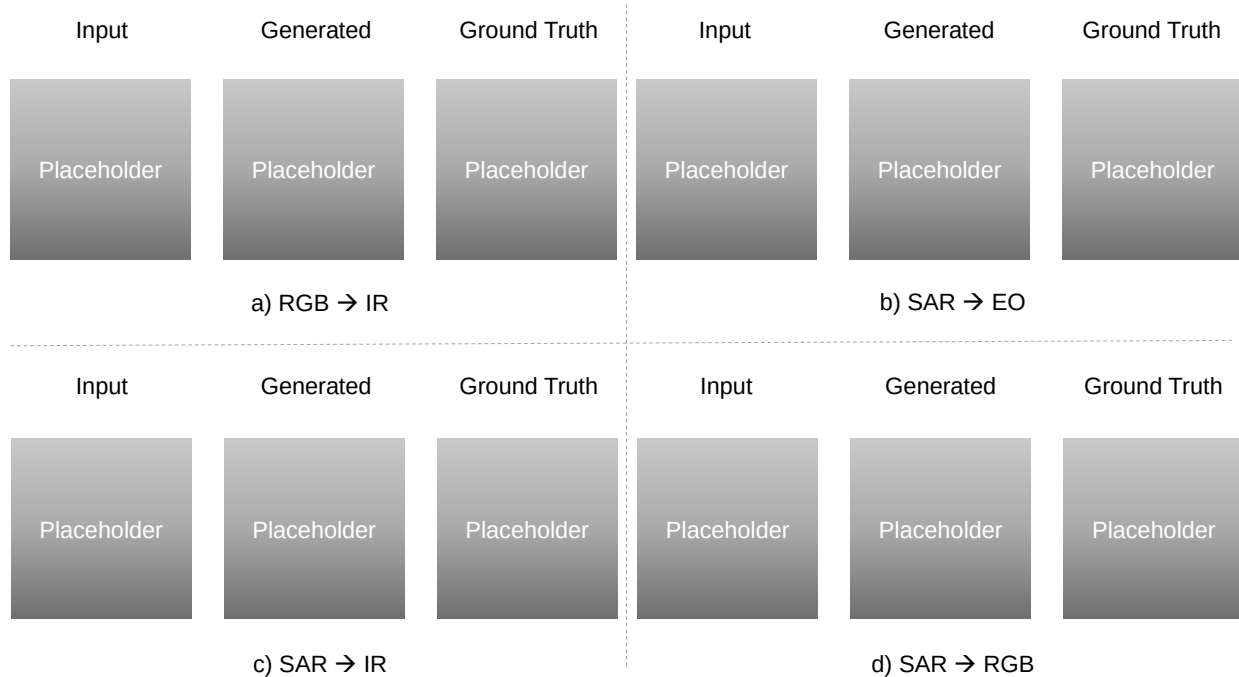


Figure 1. Teaser: The proposed EarthBridge framework for multi-modal aerial-view image translation across Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR) modalities. Our method accurately bridges the domain gap, preserving structural integrity while synthesizing high-fidelity target textures.

Abstract

001 Cross-modal image-to-image translation among Electro-
 002 Optical (EO), Infrared (IR), and Synthetic Aperture Radar
 003 (SAR) sensors is essential for comprehensive multi-modal
 004 aerial-view analysis. However, translating between these
 005 modalities is notoriously difficult due to their distinct elec-
 006 tromagnetic signatures and geometric characteristics. This
 007 paper presents **EarthBridge**, a high-fidelity translation
 008 framework developed for the 4th Multi-modal Aerial View
 009 Image Challenge – Translation (MAVIC-T). EarthBridge
 010 is founded on **Diffusion Bridge Implicit Models (DBIM)**,

which we generalize using non-Markovian bridge processes
 to enable high-quality deterministic sampling. Our method
 employs a channel-concatenated UNet denoiser trained
 with Karras-weighted bridge scalings and a specialized
 “booting noise” initialization to handle the inherent ambi-
 guity in cross-modal mappings. We evaluate EarthBridge
 across all four challenge tasks (SAR→EO, SAR→RGB,
 SAR→IR, RGB→IR), achieving superior spatial detail and
 spectral accuracy. Our solution achieved a composite score
 of **[TODO: 0.XXX]**, securing a **[TODO: Rank]** position on
 the MAVIC-T leaderboard.

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1. Introduction

The Multi-modal Aerial View Image Challenge for cross-domain image translation [1, 19, 20] is a leading challenge held in conjunction with the Perception Beyond the Visible Spectrum (PBVS) workshop from 2023 to 2025. Data collected across multiple sensing modalities—such as Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR)—provide complementary perspectives on the same scene. However, data coverage limitations and modality-specific constraints often hinder the full utilization of these diverse sources. For instance, while SAR sensors offer all-weather capabilities and can penetrate atmospheric obstructions, their imagery is complex to interpret, and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another, enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

SAR images possess unique attributes that present challenges for both human observers and vision artificial intelligence (AI) models to interpret, owing to their electromagnetic (EM) characteristics [10–13, 21].

In this paper, we propose **EarthBridge**, a diffusion-based framework for cross-modal aerial image translation built on Diffusion Bridge Implicit Models (DBIM) [43]. Our approach generalizes the diffusion bridge framework with non-Markovian bridge processes, enabling efficient deterministic sampling. By conditioning the bridge process on the source image via channel concatenation and training a UNet denoiser under Karras-weighted bridge scalings [15], EarthBridge achieves high-fidelity translation across disparate modalities. We utilize pixel-level modeling throughout to preserve the full spectral and spatial resolution of the remote sensing data.

Our findings indicate that the deterministic DBIM sampling significantly improves inference speed, achieving high-quality results in as few as [TODO: 40] sampling steps. Qualitatively, EarthBridge demonstrates a superior ability to reconstruct complex urban layouts and fine-grained textures in challenging tasks such as SAR-to-RGB, effectively bridging the domain gap. Quantitatively, our solution achieves a composite task score of [TODO: 0.XXX] for the SAR→EO task and [TODO: 0.XXX] overall, ultimately placing [TODO: Rank] in the MAVIC-T challenge leaderboard.

Our main contributions are summarized as follows:

- We propose **EarthBridge**, a novel framework for paired cross-modal aerial image translation based on Diffusion Bridge Implicit Models (DBIM), which generalizes the bridge process with non-Markovian dynamics and enables efficient deterministic sampling.
- We develop a specialized training protocol using Karras-weighted bridge scalings and pixel-level modeling, optimized for the high dynamic range and resolution requirements of multimodal remote sensing data.
- We conduct comprehensive experiments on all four MAVIC-T tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR), reporting state-of-the-art results and providing detailed quantitative and qualitative analysis.

2. Related Work

Translating across remote sensing (RS) modalities, such as SAR to optical (S2O), is critical for multisource data fusion. In this section, we provide a comprehensive literature review of image-to-image translation, diffusion bridge models, and cross-modality translation to complement the specialized discussion on remote sensing.

General Image-to-Image Translation. Image-to-image translation has been extensively studied in computer vision. Foundational works include Pix2Pix for paired conditional translation [14] and CycleGAN for unpaired translation [45]. Subsequent advances include contrastive learning for unpaired translation [24], zero-shot [25] and one-step translation with text-to-image models [26].

Diffusion Bridge Models. Unlike standard diffusion models that generate from noise, diffusion bridge models learn a stochastic process that directly connects source and target distributions, making them well-suited for paired image-to-image translation. Denoising Diffusion Bridge Models (DDBM) [44] define a variance-preserving bridge between data distributions. Related formulations include Brownian Bridge Diffusion Models (BBDM) [16] and its bidirectional extension BiBBDM [40], Image-to-Image Schrödinger Bridge (I^2SB) [17], Diffusion Schrödinger Bridge matching [34], Dual Diffusion Implicit Bridges [35], and Direct Diffusion Bridge with data consistency [2]. Recent extensions include Consistency Diffusion Bridge [6], Diffusion Bridge Implicit Models [43], deterministic translation via denoising Brownian bridge with dual approximators [39], and adaptive domain shift for cross-modality translation [38].

Cross-Modality Translation. Image translation across diverse sensor modalities—grayscale, near-infrared (NIR), thermal colorization, and color transfer—has been ad-

dressed with probabilistic color correction [23], NIR imagery colorization [36], deep colorization [21, 41].

GAN-based RS Translation. Early attempts at translating between SAR and optical modalities leveraged Generative Adversarial Networks (GANs). For instance, Wang et al. [37] proposed a hierarchical latent feature approach to improve the fidelity of SAR-to-optical translation. Hu et al. [9] specifically addressed this task in fire-disturbed regions using GANs to reconstruct optical details from SAR signals, which is vital for post-fire damage assessment when clouds or smoke obscure optical sensors.

Diffusion Models for RS Translation. Reflecting the recent shift in generative modeling, diffusion models have been increasingly applied to RS translation tasks. Qin et al. [28] introduced a conditional diffusion model incorporating spatial-frequency refinement to capture both global structure and fine-grained textures in SAR-to-optical translation. To address the computational overhead of iterative sampling, Qin et al. [29] later developed an efficient end-to-end diffusion model capable of one-step SAR-to-optical translation, significantly reducing inference latency while maintaining high generation quality. Furthermore, Wang et al. [38] proposed adaptive domain shift mechanisms within diffusion frameworks to better handle the large distribution gaps inherent in cross-modality remote sensing datasets. These advancements underscore the growing role of stochastic bridge processes and diffusion-based refinement in enabling robust, high-resolution translation across heterogeneous sensor modalities.

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y . In the context of the MAVIC-T challenge, four translation tasks are defined (see Tab. 1): SAR→EO (1 → 1 channel, 256 × 256), SAR→RGB (1 → 3, 1024 × 1024), SAR→IR (1 → 1, 1024 × 1024), and RGB→IR (3 → 1, 1024 × 1024).

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS [42] (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual

fidelity, FID [8] (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi} \arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \quad (1)$$

where the arctan normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of +1 added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

4. EarthBridge: Diffusion Bridge Implicit Models for RS Translation

4.1. Diffusion Bridge Foundation

Diffusion models have recently revolutionized generative modeling, yet they are fundamentally designed to transport complex distributions to a standard Gaussian prior. Adapting them to translate between two arbitrary distributions, such as heterogenous remote sensing modalities, requires conditioning the diffusion process on a fixed endpoint. This is achieved via Doob’s h -transform [5, 30].

Stochastic Bridges via h -transform. Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the conditioned SDE that almost surely reaches a prescribed target $x_0 = y$ at $t = 0$ starting from a source $x_T = x$ at $t = T$ is given by:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = y | x_t)dt, \quad (2)$$

where $p(x_0 = y | x_t)$ is the transition density of the original process. When the endpoints are fixed, the resulting conditioned process is called a *diffusion bridge* [3, 7, 18, 27, 32, 33], which forms the foundational mechanism for transporting one state to another. In the context of cross-modal translation, x represents the source modality (e.g., SAR) and y the target modality (e.g., optical). This formulation ensures that the generative trajectory is directly constrained by both endpoints, providing a more principled framework for paired translation than standard conditional diffusion.

Denoising Diffusion Bridge Models. Denoising Diffusion Bridge Models (DDBM) [44] provide a tractable realization of Eq. 2. Under a variance-preserving (VP) schedule, the forward bridge process constructs intermediate states z_t as:

$$z_t = a_t x + b_t y + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where a_t, b_t, σ_t are schedule-dependent coefficients. While DDBM excels in high-fidelity generation, its inference typically relies on simulating Eq. 2 via computationally expensive numerical solvers (e.g., Heun or stochastic samplers), often requiring hundreds of function evaluations (NFE) for optimal results. In this work, we adopt the more efficient *implicit* formulation to accelerate this process.

4.2. EarthBridge: Non-Markovian DBIM

We present our EarthBridge approach based on **Diffusion Bridge Implicit Models (DBIM)** [43], which generalizes the diffusion bridge framework with non-Markovian bridge processes defined on discretized timesteps. Our solution leverages the deterministic sampling property of DBIM to achieve high-fidelity cross-modal translation with significantly fewer sampling steps compared to the vanilla DDBM. Fig. 2 shows an overview of our EarthBridge pipeline.

Non-Markovian Diffusion Bridges. In contrast to the continuous-time Markovian bridge processes in DDBM, we consider a family of non-Markovian diffusion bridges defined over a sequence of discretized timesteps $0 = t_0 < t_1 < \dots < t_N = T$, controlled by a variance parameter $\rho \in \mathbb{R}^{N-1}$. The joint probability distribution $q^{(\rho)}(\mathbf{z}_{t_0:N-1} | \mathbf{x})$ is factorized through a chain of conditional distributions:

$$q^{(\rho)}(\mathbf{z}_{t_0:N-1} | \mathbf{x}) = q_0(\mathbf{z}_{t_0}) \prod_{n=1}^{N-1} q^{(\rho)}(\mathbf{z}_{t_n} | \mathbf{z}_0, \mathbf{z}_{t_{n+1}}, \mathbf{x}), \quad (4)$$

where q_0 is the target modality distribution and each step $q^{(\rho)}$ is defined as:

$$\mathcal{N}(a_{t_n} \mathbf{x} + b_{t_n} \mathbf{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \boldsymbol{\eta}_{n+1}, \rho_n^2 \mathbf{I}), \quad (5)$$

with $\boldsymbol{\eta}_{n+1} = (\mathbf{z}_{t_{n+1}} - a_{t_{n+1}} \mathbf{x} - b_{t_{n+1}} \mathbf{z}_0) / c_{t_{n+1}}$. Here, a_t, b_t, c_t are known coefficients determined by the VP log-SNR schedule (Sec. 4.1). Under this construction, the marginal distributions $q^{(\rho)}(\mathbf{z}_{t_n} | \mathbf{x})$ are preserved to be consistent with the continuous-time forward bridge (Eq. (3)). The variance parameter ρ controls the level of stochasticity in the sampling process, ranging from a purely Markovian stochastic process to a fully deterministic implicit model.

Reverse Generative Process and Sampling. The generative process is initialized at the source image $\mathbf{z}_T = \mathbf{x}$ and iteratively samples $\mathbf{z}_{t_n}$ from $\mathbf{z}_{t_{n+1}}$ via a parameterized data predictor $D_\theta(\mathbf{z}_{t_{n+1}}, t_{n+1}, \mathbf{x})$. The core updating rule is given by:

$$\mathbf{z}_{t_n} = a_{t_n} \mathbf{x} + b_{t_n} \hat{\mathbf{z}}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \hat{\boldsymbol{\epsilon}} + \rho_n \boldsymbol{\epsilon}, \quad (6)$$

Table 1. Task specifications and training data for the MAVIC-T challenge. Crop augmentation uses 1024×1024 sliding windows with stride 512 (50% overlap); SAR→optical pairs are georeference-aligned.

Task	Resolution	Channels	Samples (Orig.)	Samples (Aug.)
SAR→EO	256×256	1→1	89,411	–
SAR→RGB	1024×1024	1→3	10,576	75,703
SAR→IR	1024×1024	1→1	10,576	74,317
RGB→IR	1024×1024	3→1	2,272	7,388

where $\hat{\mathbf{z}}_0 = D_\theta(\mathbf{z}_{t_{n+1}}, t_{n+1}, \mathbf{x})$ is the clean data prediction and $\hat{\boldsymbol{\epsilon}}$ is the predicted noise at time t_{n+1} . Setting $\rho_n = 0$ for all steps yields the deterministic *Diffusion Bridge Implicit Model (DBIM)*, which can be viewed as an Euler discretization of a probability flow ODE specifically tailored for diffusion bridges.

Bridge Scalings. Following the Diffusion Bridge framework, the raw model output F_θ is post-processed through bridge scalings ($c_{\text{skip}}, c_{\text{out}}, c_{\text{in}}$) derived from the VP log-SNR schedule. The denoised prediction is defined as:

$$D_\theta(\mathbf{z}_t, t, \mathbf{x}) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot \mathbf{z}_t, t, \mathbf{x}) + c_{\text{skip}}(t) \cdot \mathbf{z}_t, \quad (7)$$

Booting Noise. A critical refinement in our sampling procedure is the introduction of *booting noise* at the initial step ($t_N = T$). Since $c_{t_N} = 0$, the denominator in $\hat{\boldsymbol{\epsilon}}$ leads to singularity. To address this, we force the initial step to be stochastic by choosing $\rho_{N-1} = c_{t_{N-1}}$, which injects a standard Gaussian booting noise $\boldsymbol{\epsilon}_{\text{boot}}$. This booting noise serves as the latent variable, encoding the one-to-many mapping ambiguity in RS translation (e.g., several plausible infrared images for a single SAR input).

5. Experiments

5.1. Dataset and Preprocessing

We use the MAVIC-T 2025 dataset from the challenge organizers, with imagery from UNICORN, USGS EROS HRO, and UMBRA across four regions: Bingham, Centerfield, Manhattan, and UC Davis. Tab. 1 summarizes the four translation tasks. SAR→EO operates at 256×256 ; the other three tasks use 1024×1024 . For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. We also create crop-augmented training sets for the 1024×1024 tasks to increase data diversity.

5.2. Architecture and Training Objectives

UNet Denoising Backbone Our denoiser D_θ is a UNet [31] that predicts the clean target from the noisy sample $\mathbf{z}_t$ conditioned on timestep t and source $\mathbf{x}$. We use

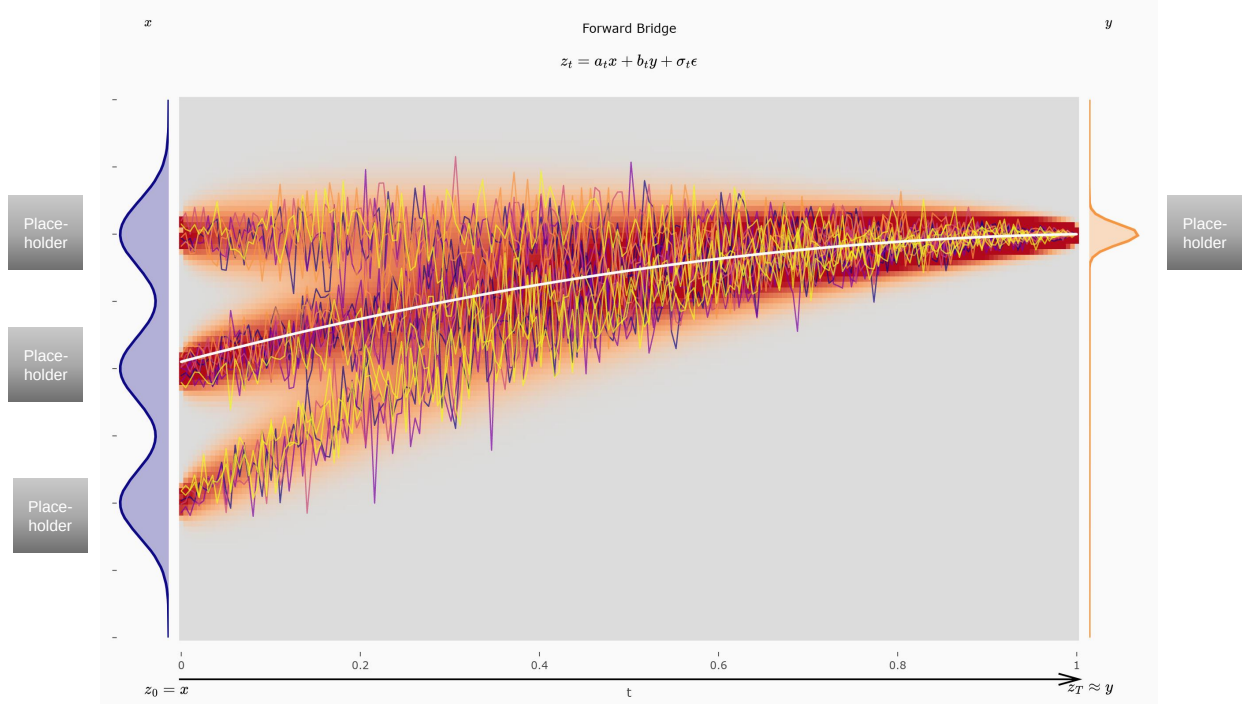


Figure 2. Overview of our EarthBridge translation pipeline based on DBIM. The source image x is concatenated with the noisy sample z_t and fed to a UNet denoiser. During training, the bridge forward process constructs z_t from the target y and source x via bridge scalings. During inference, deterministic DBIM sampling transforms x into the translated output $\hat{z}_0$.

task-specific scaling (128 base channels with resolution-dependent multipliers) and train with the Prodigy [22] optimizer. Detailed architectural parameters and source conditioning strategies are provided in Sec. 6.1.

Training Objective The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(z_t, t, x)$ and the clean target y :

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(x, y), t, \epsilon} [w(t) \cdot \|D_\theta(z_t, t, x) - y\|_2^2], \quad (8)$$

where $w(t)$ is the Karras bridge weighting. The diffusion time t is sampled from a Karras inverse- ρ distribution. During inference, we utilize deterministic DBIM sampling for efficiency. Full technical details on the loss weighting, time-sampling distribution, and per-task training configurations are deferred to the appendix (Sec. 6).

5.3. Quantitative Results

Tab. 2 presents the main quantitative performance of EarthBridge across the challenge tasks. The composite task score provides a balanced evaluation of perceptual, distributional, and structural fidelity. EarthBridge demonstrates consistent performance across both low- and high-resolution tasks.

5.4. Qualitative Results

Fig. 3 shows qualitative examples across all four tasks. EarthBridge preserves fine-grained structural information from the source modality while accurately synthesizing target textures, demonstrating consistent performance across both low- and high-resolution settings.

5.5. Failure Case and Challenging Scenario Analysis

[TODO: Add failure case analysis and challenging scenarios.]

6. Implementation Details

This section provides the comprehensive technical details of the EarthBridge architecture, training hyperparameters, and multi-modal refinement modules.

6.1. Architecture, Sampling, and Scheduler Settings

The EarthBridge architecture and sampling configurations are designed for high-fidelity cross-modal translation with efficiency. We utilize a UNet2DModel backbone (ADM-style [4]) with per-task architectural parameters: number of channels, number of residual blocks, attention resolutions, channel multipliers, resolution, and training settings

Table 2. Main results: EarthBridge quantitative performance across the four MAVIC-T translation tasks. We report LPIPS ($\downarrow$), FID ($\downarrow$), L1 ($\downarrow$), and the composite Task Score ($\downarrow$).

Method	SAR→EO				SAR→RGB				SAR→IR				RGB→IR				Overall
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	
EarthBridge (Ours)	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]	[0.0]



Figure 3. Qualitative results across all four MAVIC-T translation tasks. Each row corresponds to one task (SAR→EO, SAR→RGB, SAR→IR, RGB→IR). Columns show: source image, EarthBridge output, and ground-truth target. Our model preserves structural layout from the source while synthesizing faithful target-modality textures across both 256×256 and 1024×1024 resolutions.

(epochs, augmentations) as detailed in Tab. 3. Timestep information is injected via sinusoidal embeddings and Adaptive Group Normalization (AdaGN). Source image conditioning is implemented via channel-wise concatenation (concat mode) $[z_t, x]$.

Sampling is performed using the deterministic DBIM formulation [43] with 1000 inference steps and a Gaussian booting noise ϵ_{boot} at $t = T$. The bridge process follows a variance-preserving (VP) log-SNR schedule with $\beta_d = 2.0$ and $\beta_{\text{min}} = 0.1$. For training, we sample diffusion time from a Karras inverse- ρ distribution with $\rho = 7$. Per-task architectural and training configurations are summarized in Tab. 3.

We train EarthBridge using the Prodigy adaptive optimizer across all translation tasks. The training objective follows the weighted Karras bridge loss with task-specific batch sizes and augmentations.

7. Lessons and Insights

We summarize key lessons and insights from developing EarthBridge for the MAVIC-T translation track.

Insight 1: Diffusion Bridge vs. Standard Diffusion

The diffusion bridge directly builds a mapping from the source domain to the target domain, whereas standard diffusion starts from a noise prior (typically Gaussian noise) and requires a conditional generation strategy to accomplish this task. We hypothesize that the diffusion bridge is more efficient for training and achieves better results for cross-modal translation. However, standard diffusion can be more flexible for further conditional development, e.g., leveraging pre-trained models for a good prior. There is a trade-off between these approaches; we encourage readers to further explore both directions.

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Table 3. Per-task EarthBridge architectural and training configurations. Resolution denotes the training image input resolution.

Parameter	SAR→EO	SAR→RGB	SAR→IR	RGB→IR
Number of channels	64	96	96	128
Number of residual blocks	2	2	2	2
Attention resolutions	N/A	32, 16, 8	32, 16, 8	32, 16, 8
Attention head dimension	N/A	64	64	64
Resolution	256×256	512×512	512×512	512×512
Input channels	2	1	2	4
Output channels	1	3	1	1
Channel multipliers	(1,2,3,4)	(1,1,2,2,4,4)	(1,1,2,2,4,4)	(1,1,2,2,4,4)
Number of epochs	55	24	39	86
Use augmented	N/A	✓	✓	✓
Use random crop	N/A	✓	✓	✓
Use horizontal flip	✓	✓	✓	✓
Use vertical flip	✓	✓	✓	✓

Insight 2: Resolution Scaling and Multi-Stage Curriculum

For 1024px modeling, we train on 512px only and directly infer on 1024px, which yields strong results while saving training compute budget. If time permits, we may further perform a quick fine-tuning on 1024px directly—we term this approach *multi-stage resolution curriculum learning*.

Insight 4: Inference Steps and Metric Trade-offs

Using the DBIM ODE solver, few-step inference yields visually appealing results. For RGB→IR, even a single-step generation is sufficient, and increasing steps tends to hallucinate details. In practice, however, many more inference steps greatly improve FID and LPIPS in the competition metrics, even when L1 remains unchanged—highlighting a trade-off between perceptual quality, metric optimization, and inference efficiency.

Insight 3: Task Difficulty and Input Modality

We found that SAR→EO and RGB→IR are substantially easier than the other two tasks. For RGB→IR, the three-channel input provides sufficient information for the translation. For SAR→EO, the relative ease may stem from both modalities originating from the same sensor, reducing the domain gap.

8. Lessons and Insights

We summarize key lessons and insights from developing EarthBridge for the MAVIC-T translation track.

9. Conclusion

In this paper, we presented EarthBridge, a novel solution for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T). By leveraging Diffusion Bridge Implicit Models (DBIM), we generalized the diffusion bridge framework with non-Markovian processes, enabling efficient and deterministic cross-modal translation. Our approach effectively handles disparate sensor characteristics across EO, IR, and SAR modalities through pixel-level modeling and a specialized training protocol featuring Karras-weighted bridge scalings. Quantitative and qualitative evaluations

371 across four translation tasks demonstrate EarthBridge’s su-
372 perior ability to preserve spatial structure while generating
373 high-fidelity target textures. Future work will explore the
374 integration of adaptive domain shift mechanisms and fur-
375 ther optimization for real-time remote sensing applications.
376 **[TODO: Add final ranking and summary of key find-**
377 **ings.]**

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Supplementary Material